No Evidence of Cloud Seeding Signatures in the NEXRAD Radar Record

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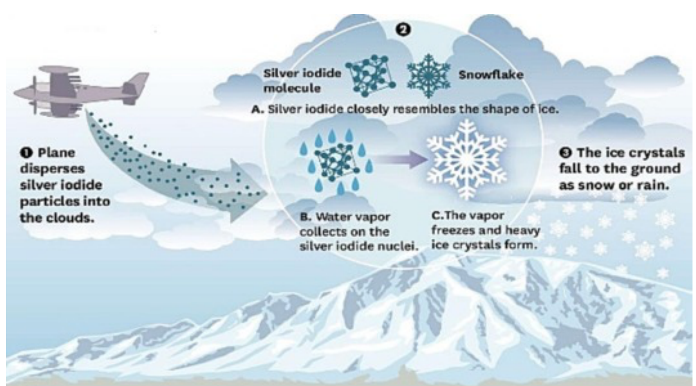
Abstract

Cloud seeding has been practiced operationally for decades, yet evidence for its efficacy remains difficult to isolate due to strong natural variability and non-random operational targeting. This project investigates whether cloud seeding operations leave detectable signatures in the NEXRAD radar record using an unsupervised anomaly detection framework. Composite NEXRAD reflectivity mosaics were processed into 256×256 pixel tiled radar scenes and encoded using a convolutional autoencoder. Temporal evolution in latent space was then modeled using a causal transformer trained on non-event radar sequences. The framework was evaluated on two contrasting case studies: the SNOWIE winter orographic cloud seeding experiment in Idaho and the Trans-Pecos Weather Modification Association (TPWMA) operational convective seeding program in West Texas. The k-NN anomaly scorer achieved AUROCs of 0.841 (SNOWIE) and 0.742 (Rainmaker); the transformer reached AUROC 0.9605 on Rainmaker test sequences. Despite this separation, results do not provide clean evidence of cloud-seeding-specific signatures in the NEXRAD record. SNOWIE suggests that physically real seeding plumes may exist below NEXRAD detection limits, while the Rainmaker results highlight persistent attribution challenges due to operational targeting of already-favorable convective environments.

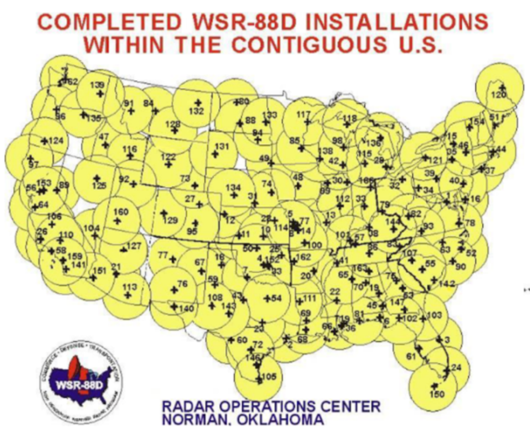
Motivation



(Shafiei et al. 2023)

Cloud seeding has been practiced operationally for nearly eighty years, yet evidence for its efficacy remains difficult to isolate and interpret. Modern cloud seeding programs are active throughout the western United States and many other regions worldwide, typically with the goal of increasing precipitation or enhancing snowpack through the dispersal of silver iodide or hygroscopic particles into suitable cloud systems (GAO, 2024; National Academies, 2015). Silver iodide acts as an ice-nucleating agent because its crystal lattice structure closely resembles that of hexagonal ice, promoting the freezing of supercooled water droplets at temperatures warmer than would otherwise occur (Shafiei et al., 2023). While some studies—particularly winter orographic experiments such as SNOWIE—have produced physical evidence supporting localized microphysical effects (Tessendorf et al., 2019), broader questions surrounding operational effectiveness remain unresolved.

A major challenge is that cloud seeding operations are rarely conducted under randomized or controlled conditions. Aircraft are typically dispatched into storms that already appear meteorologically favorable, making it difficult to separate potential seeding effects from natural storm evolution. Radar provides a possible avenue for evaluating these operations at larger spatial and temporal scales. The NEXRAD WSR-88D network offers continuous, high-frequency coverage across much of the contiguous United States and archives decades of radar observations (NCEI, n.d.). If cloud seeding meaningfully alters storm structure or precipitation organization at the radar scale, those changes may appear as statistically anomalous patterns within the radar record.



(Brodzik 2020)

At the same time, radar observations impose important physical limitations. Radar reflectivity is strongly weighted toward larger hydrometeors due to the approximate relationship $Z \propto D^6$, meaning that a small number of large drops or hailstones can dominate the radar signal over many smaller ice particles. As a result, weak or shallow cloud seeding signatures—such as the ice crystal plumes documented during SNOWIE—may remain below the detection threshold of operational radar systems even if microphysical changes are occurring within the cloud.

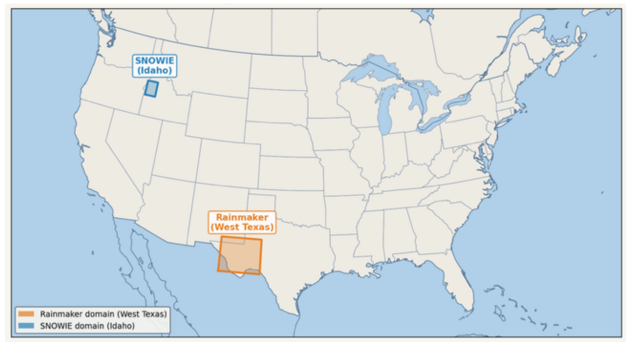
This study investigates whether operational cloud seeding events leave detectable signatures in NEXRAD composite reflectivity mosaics using an unsupervised machine learning framework. Rather than attempting direct causal attribution, the project evaluates whether seeded events appear statistically unusual relative to normal radar evolution across two contrasting study regions and seeding regimes.

Methods

Datasets

The primary radar dataset consisted of NEXRAD composite reflectivity mosaics obtained through the Iowa Environmental Mesonet (Iowa Environmental Mesonet, n.d.). Composite reflectivity products represent the maximum reflectivity observed through the vertical radar column, providing broad spatial coverage and consistent visualization of storm organization across large regions. Radar reflectivity values were encoded as uint8 and converted to dBZ using the standard Iowa State relationship (dBZ = pixel/2 − 32).

Two cloud seeding case studies were examined. The first was the Seeded and Natural Orographic Wintertime Clouds: The Idaho Experiment (SNOWIE), a winter field campaign conducted in the Payette River Basin, Idaho during 2016–2017 (Tessendorf et al., 2019). SNOWIE was one of the first cloud seeding experiments in the United States to employ a randomized treatment design alongside close-range Doppler on Wheels (DOW) radar observations. This combination enabled direct documentation of silver iodide seeding plume structures—elongated bands of enhanced reflectivity downwind of the seeding aircraft—providing a physically verified example of a detectable cloud seeding effect (Tessendorf et al., 2019; UCAR EOL, n.d.).



The second dataset came from the Trans-Pecos Weather Modification Association (TPWMA), a cloud seeding contractor operating in the Trans-Pecos region of West Texas. TPWMA conducts warm-season operational convective cloud seeding using aircraft-mounted glaciogenic and hygroscopic flares under contract with agricultural and ranching interests in the region (TPWMA, n.d.). Unlike SNOWIE, TPWMA operations are purely operational—seeding decisions are made in real time based on meteorological conditions, with no randomized controls or scientific instrumentation beyond standard flight documentation. TPWMA operations records, including flight timing, flare deployment counts, operational notes, and target counties, were compiled from publicly available Trans-Pecos operations reports (TPWMA, n.d.), hereinafter referred to as "Rainmaker." ERA5 700 hPa wind fields from the Copernicus Climate Change Service (2018) were incorporated to assess upper-level steering flow and storm motion during selected events.

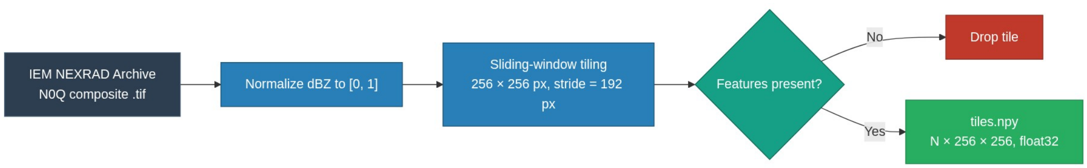
Data Collection

Radar mosaics were downloaded as GeoTIFF imagery spanning both background non-event periods and targeted seeding windows. Background periods were selected to represent ordinary radar evolution within each study region and served as training data for the anomaly detection framework. Event windows corresponding to SNOWIE intensive observation periods (IOPs) and TPWMA seeding operations were reserved for testing and evaluation. Post-April 2022 composites are served at 0.25 km resolution (N0B product); these were automatically detected by pixel size and 2× average-pooled to match the 0.5 km resolution of earlier composites, ensuring a consistent input resolution across the full time series.

Data Processing

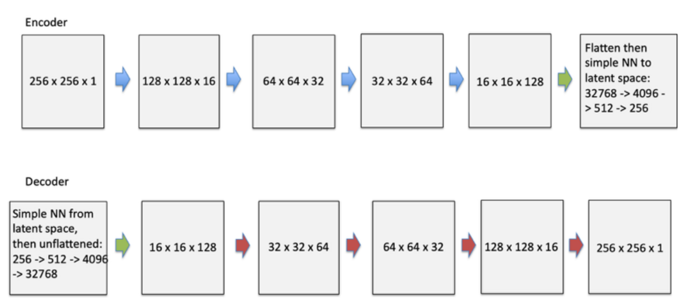
Radar mosaics were spatially tiled into 256×256 pixel patches using a sliding window with stride 192 pixels (25% overlap), creating localized radar scenes suitable for machine learning analysis. Tiles with a maximum normalized reflectivity below 0.48 (approximately 20 dBZ) were discarded as clear-sky frames, further runs should lower this threshold (to about 5 dBZ). Training and testing datasets were separated temporally to prevent leakage between ordinary background frames and seeding event periods.

For the Rainmaker (TPWMA) study region, approximately 907,883 training tiles and 1,701 testing tiles were generated across 46 operational seeding days. For the SNOWIE region, approximately 3,860,976 training tiles and 7,045 testing tiles were produced across 24 intensive observation periods.



Autoencoder

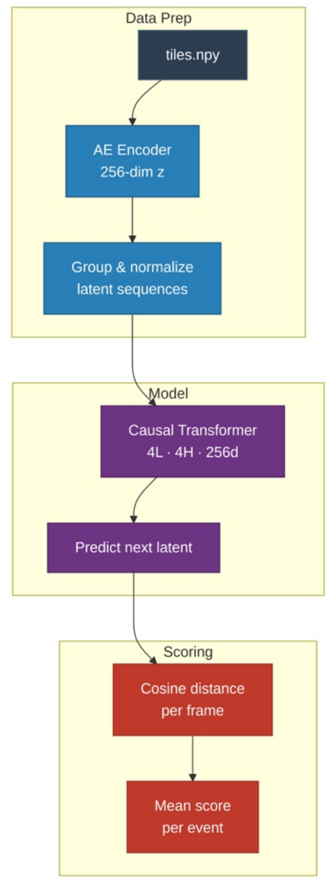
A convolutional autoencoder was trained to learn compact latent representations of ordinary radar structure. The encoder compressed each 256×256 radar tile through four convolutional blocks (1→16→32→64→128 channels, each followed by 2x2 max-pooling) and a three-layer MLP into a 256-dimensional latent vector. The decoder reconstructed the original tile from this latent representation via transposed convolution layers. Training used MSE loss and the Adam optimizer on background (non-seeding) frames only, so the model learned to represent ordinary radar evolution without any labeled examples of cloud seeding effects. The latent vectors produced by the encoder serve as compact summaries of radar structure that can be compared and scored.



This is the model architecture for the autoencoder. Each blue arrow represents a convolution + pooling, and each red arrow represents a convolutional upsample. The green arrows represent flattening or unflattening to or from a convolutional layer, and those cells represent multiple 1d layers.

k-NN Anomaly Scoring

Per-tile anomaly scores were computed using k-nearest-neighbor (k-NN) cosine distance in the normalized latent space. A FAISS index was built over all training-set latent vectors; each tile's score is the mean cosine distance to each neighbor in the reference set. In order to properly simulate true OOS connectivity, the base distribution was built within the training sample, with 100 graphs being created, then tested with data from the training set not included in the graph. Then, for the test set, the same structure was utilized, which data being randomly removed from the training set to emulate the same size dataset during the training graph creation. The net result is that p-values can be drawn from the anomaly detection, without knowing the true underlying distribution. Tiles whose radar structure is unlike the training distribution—including seeding-event frames—receive higher anomaly scores, with p-values ranging from .03-.06.



Causal Transformer

To model temporal radar evolution, latent vectors were organized into per-tile-position time series and used to train a causal transformer—a decoder-only architecture in the style of Vaswani et al. (2017)—with 4 layers, 4 attention heads, and 256-dimensional embeddings. The transformer received 12 consecutive latent vectors (one hour of 5-minute scans) as context and was trained to predict the 13th. Anomaly score at inference time is the cosine distance between the transformer's prediction and the actual observed latent vector. Larger cosine distances indicate radar evolution that deviates substantially from learned temporal patterns.

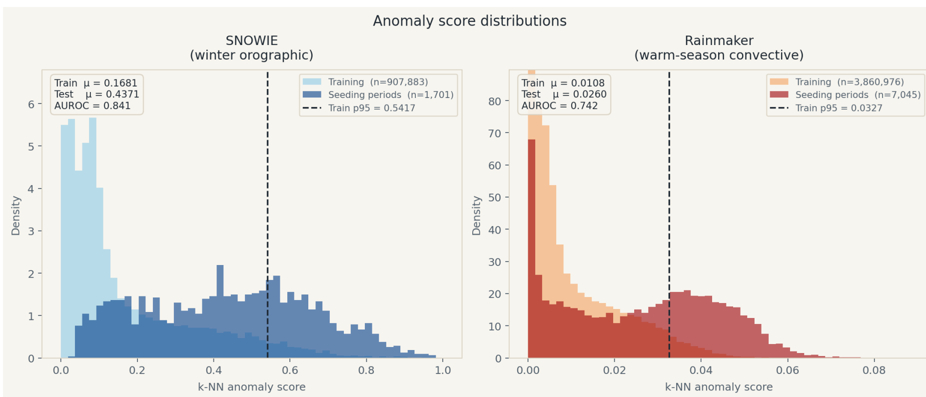
Code

All data processing, tiling, model training, and analysis were implemented in Python using PyTorch, rasterio, NumPy, and FAISS. Pipeline scripts and documentation are archived in the project directory on the UChicago Midway3 computing cluster.

Results

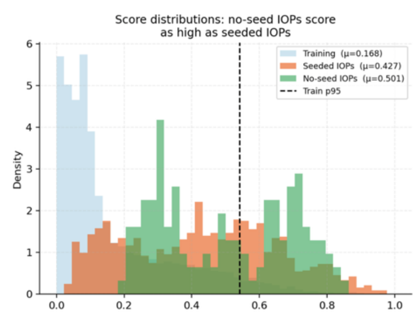
Autoencoder Performance

The autoencoder successfully reconstructed ordinary radar scenes across both study regions, indicating that the model learned meaningful representations of background radar structure. Reconstruction outputs preserved broad storm morphology, convective organization, and reflectivity gradients while smoothing some fine-scale features. k-NN anomaly scoring produced clearly distinguishable score distributions between background training tiles and seeding-event test tiles.



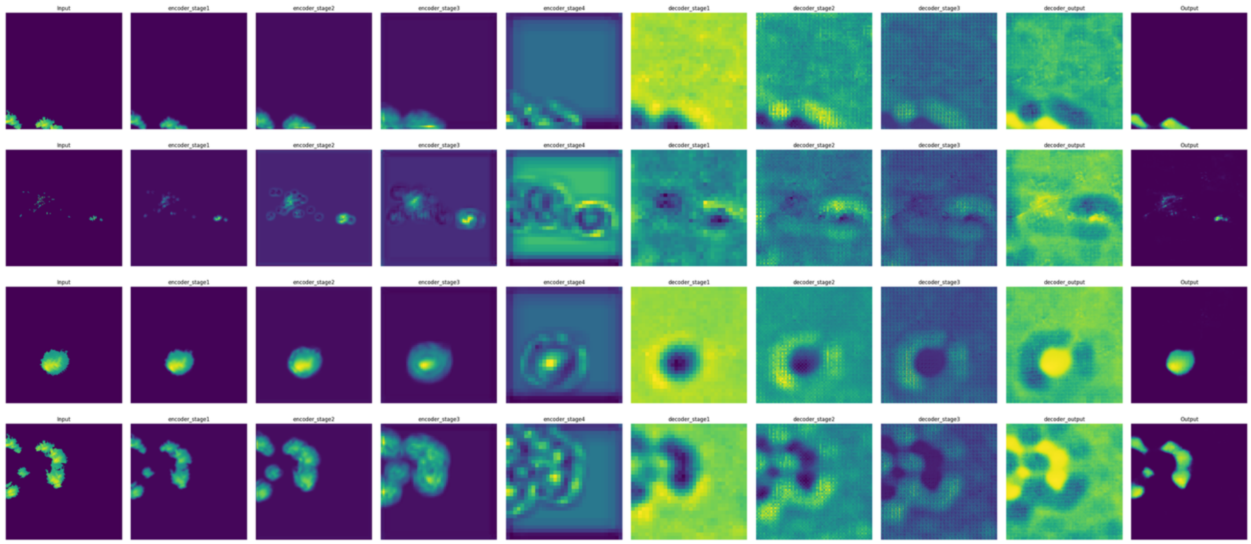
Shown here is training anomaly score histogram underneath test anomaly score histogram, with SNOWIE on LHS and Rainmaker on RHS. Both test cases had significantly different distributions, with higher mean anomaly scores, with moderate p-values.

The full k-NN score distributions across all SNOWIE IOPs, separated by seeding status. Both seeded and unseeded IOPs score substantially above the background training distribution (μ=0.168), with the seeded IOP mean reaching 0.427 and the no-seed IOP mean reaching 0.501. Notably, the unseeded control IOPs score *higher* on average than the seeded IOPs—the opposite of what would be expected if the model were detecting seeding-induced changes. This result suggests that the elevated scores across all IOPs reflect unusual storm morphology present regardless of seeding activity. The no-seed IOPs that score highest are likely events characterized by strong natural convective organization—the same conditions that can make seeding unnecessary or unsafe—rather than any signal attributable to silver iodide dispersal. Together, these distributions indicate that while the autoencoder successfully distinguishes IOP conditions

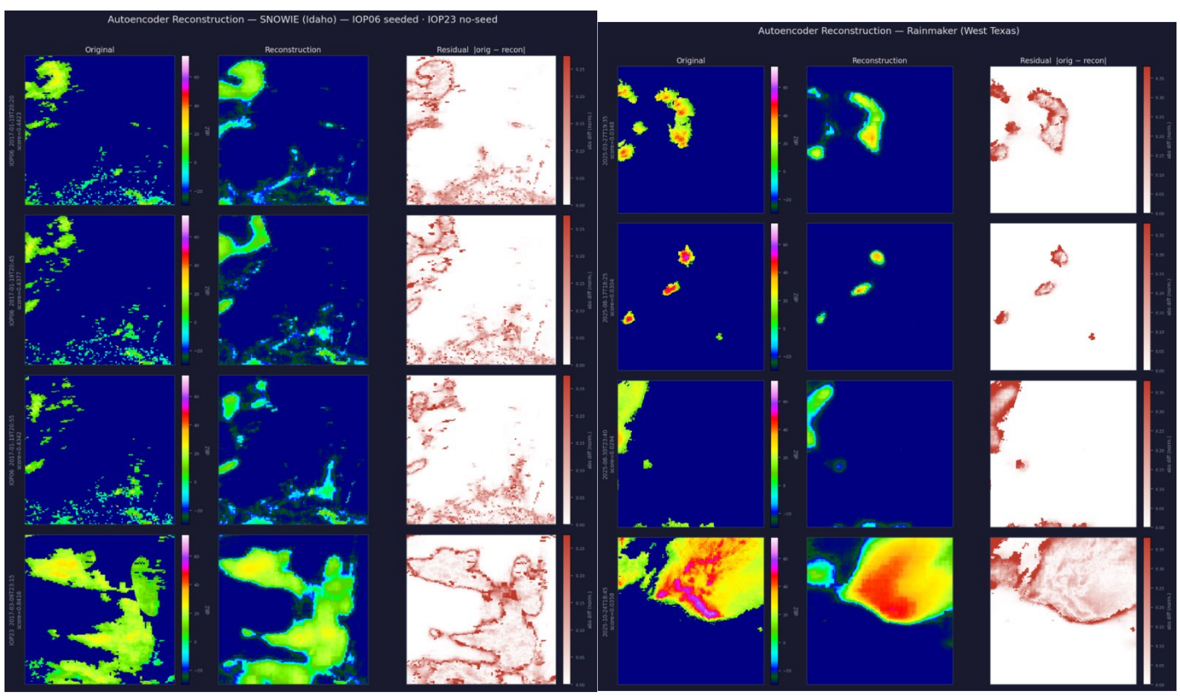


Shown here are three histograms layered on top of each other, all with SNOWIE data. Training set, non-seeded IOPs, and seeded IOPs. Although both types of IOPs had higher than base anomaly scores, they are relatively indistinguishable from each other.

from ordinary background radar evolution (AUROC = 0.841), it cannot discriminate between seeded and unseeded events within the IOP population.



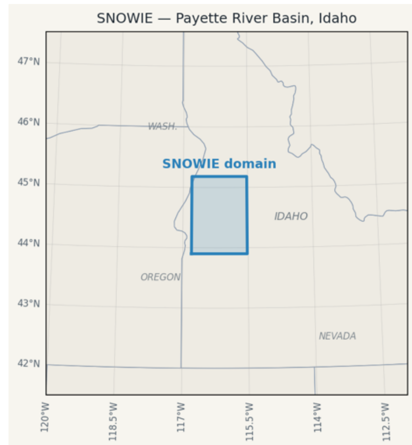
Displayed here is a demonstration of the autoencoder pipeline on 4 samples, with each sample taking a row. Each column represents a mean of all the slices in that layer, and the flattened NN in the middle of the autoencoder is not included, only the 2d a.



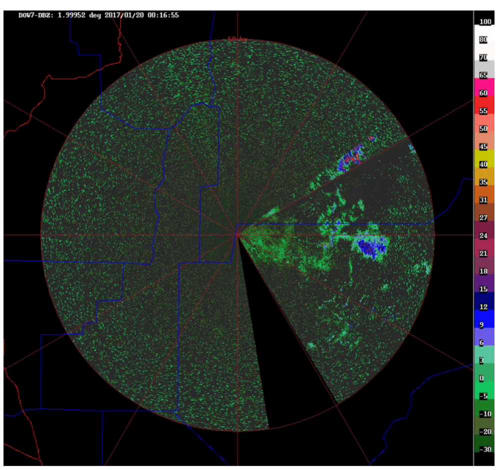
Shown here are 8 reconstruction examples. From left to right we have input (raw radar), output (decoded latents), and difference. We can see that the reconstruction softens edges and smooths centers, but retains general intensity and structure.

SNOWIE

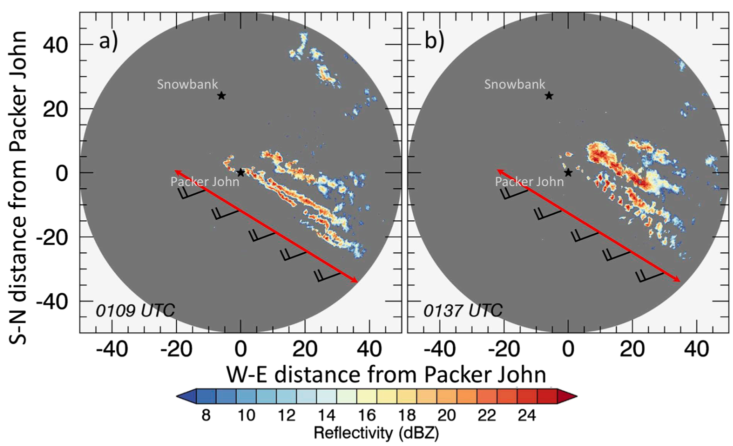
Within the SNOWIE dataset, the model produced contrasting results between intensive observation periods. IOP06, which included a documented seeded plume observed by DOW radar (Tessendorf et al., 2019), did not produce especially strong anomaly scores within the composite NEXRAD record. IOP23—an unseeded control mission—produced substantially stronger anomaly responses. Radar imagery during IOP23 displayed stronger natural storm organization and reflectivity structure than the seeded IOP06 case, indicating that the framework was responding primarily to unusual storm morphology rather than specifically to cloud seeding activity.



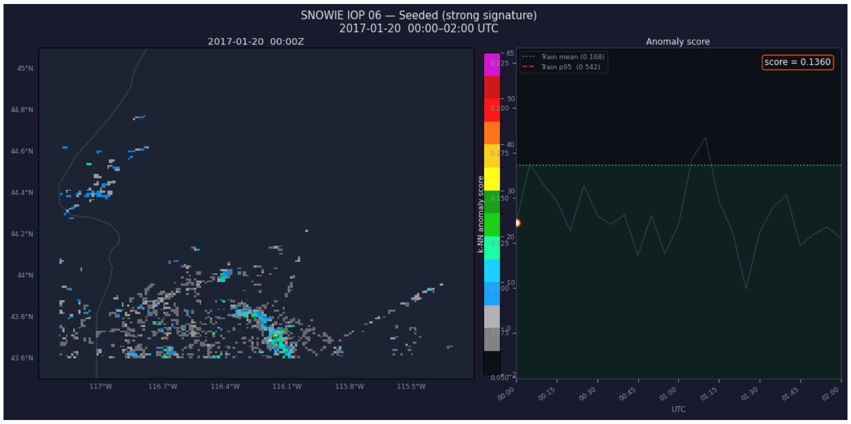
These results highlight a physical limitation inherent to this approach. SNOWIE seeded plume structures were composed primarily of relatively weak ice crystal signatures at low altitudes (Tessendorf et al., 2019). Beam overshoot, terrain blockage, and the strong reflectivity weighting toward larger hydrometeors ($Z \propto D^6$) likely reduced or eliminated plume visibility within NEXRAD composite products.

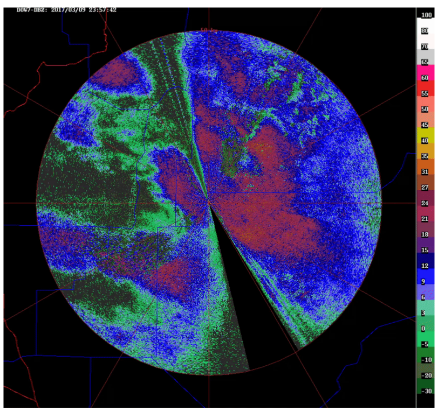


(UCAR EOL)



(Tessendorf et al. 2019)





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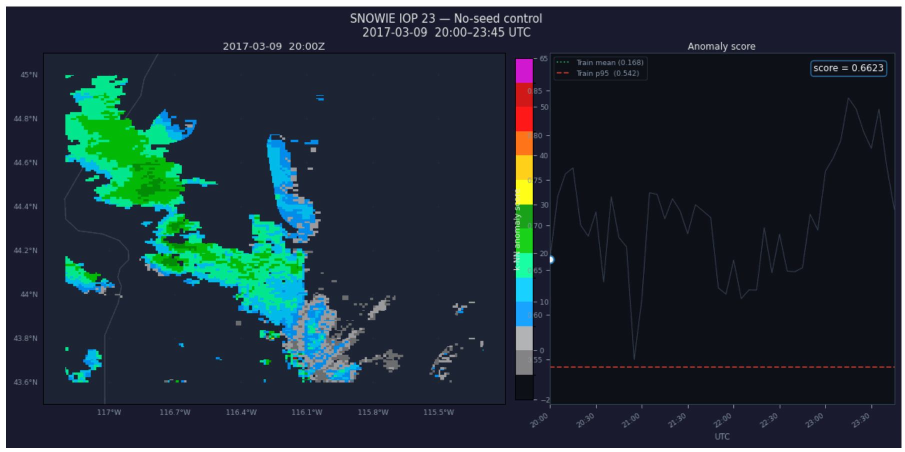
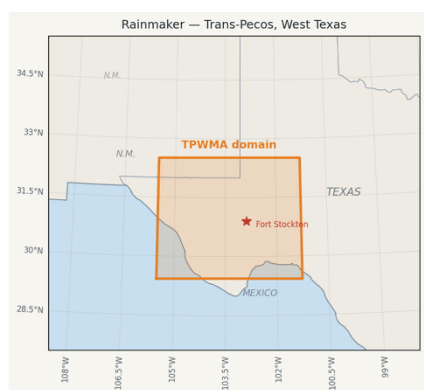
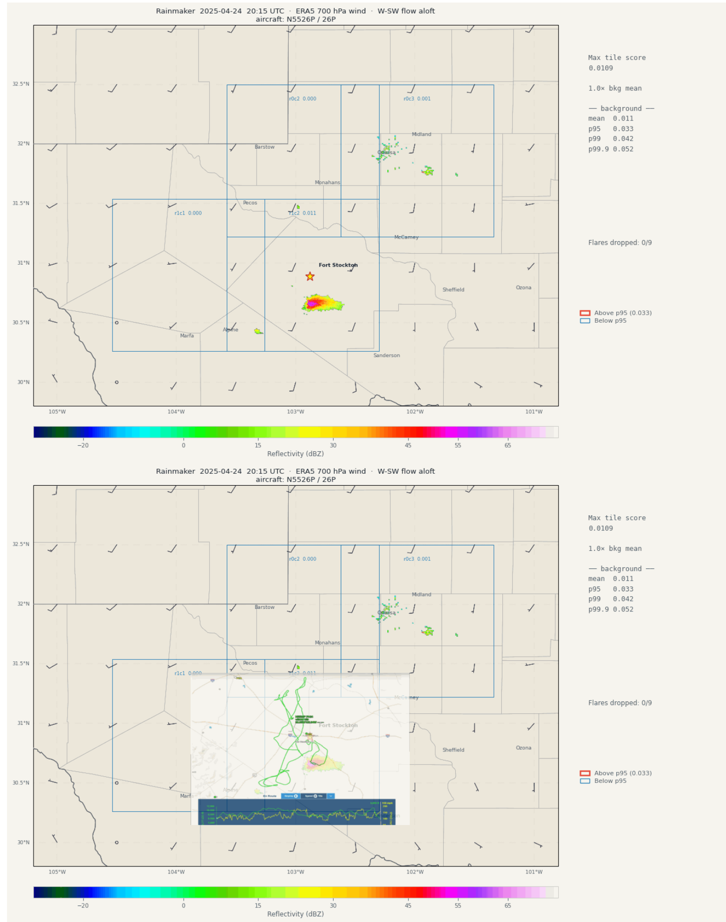


Rainmaker (TPWMA)



The Rainmaker dataset provided a more favorable physical environment for radar-based detection: flatter terrain, closer radar geometry, and deeper convective storm structure. Out of 46 operational seeding days, four events were identified as structurally anomalous and selected for detailed review: April 24, May 4, May 24, and October 8, 2025. All four exceeded background p99.9 by a factor of 5-6× in peak k-NN score. The causal transformer, scoring on one-hour latent sequences, achieved AUROC 0.9605 on the 85 available Rainmaker test sequences—a substantially stronger separation than the k-NN approach alone.

Several flagged events displayed unusual storm evolution, rapid convective intensification, or organized radar structure during or near seeding operations. However, operational records from TPWMA (n.d.) consistently showed that aircraft were dispatched into storms that already exhibited favorable convective development prior to flare deployment. In some cases, severe thunderstorm watches or strong reflectivity signatures were already present before seeding began, and additional convection developed in unseeded neighboring regions during several events.

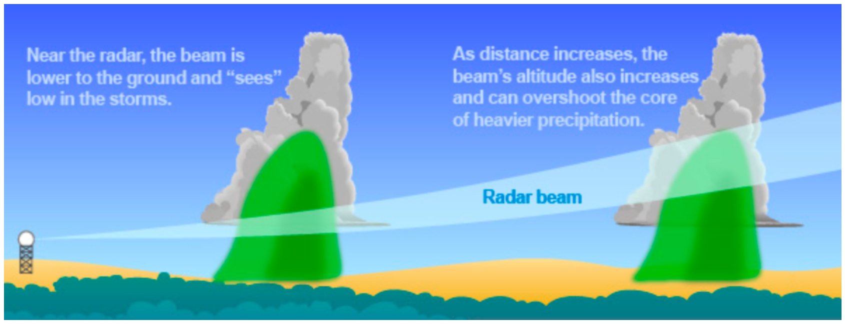


Discussion

The results demonstrate both the utility and the limitations of unsupervised radar anomaly detection for evaluating cloud seeding operations. The framework successfully learned ordinary radar evolution and identified storms with unusual structural or temporal characteristics without requiring labeled examples of seeding effects—functioning effectively as a screening and prioritization tool for identifying meteorologically interesting events within large radar archives.

At the same time, the study highlights two major constraints on radar-based cloud seeding evaluation: physical detectability limits and attribution limits.

The SNOWIE results illustrate the detectability problem. Seeded plumes documented by close-range DOW radar (Tessendorf et al., 2019) were composed of shallow, low-reflectivity ice crystal bands that are likely below the effective sensitivity of operational NEXRAD systems at the range distances involved. Even when cloud seeding produces physically real microphysical effects, those effects may not be detectable in composite NEXRAD mosaics due to beam overshoot and the $Z \propto D^6$ reflectivity weighting.



(NOAA JetStream)

The Rainmaker results illustrate the opposite problem: attribution. Even when storms are clearly detectable and structurally anomalous, TPWMA seeding decisions are strongly tied to pre-existing favorable atmospheric conditions. Flight logs from TPWMA (n.d.) repeatedly documented aircraft repositioning toward already-intensifying convection and waiting for cells to strengthen before flare deployment. This operational targeting means that the same conditions motivating seeding are also responsible for the subsequent anomalous radar evolution, making causal attribution extremely difficult even with strong anomaly scores.

The transformer's higher AUROC (0.9605 vs. 0.742 for k-NN) likely reflects its sensitivity to temporal evolution patterns rather than static frame structure—seeding-event periods may tend to exhibit more unusual storm development trajectories than single-frame k-NN scores can capture.

These findings suggest that the absence of a clear NEXRAD-scale signature should not be interpreted as proof that cloud seeding has no physical effect. Future work could improve upon this framework through the incorporation of closer-range mobile radar observations, randomized operational controls, or explicit causal inference methods.

Conclusion

This study developed an unsupervised anomaly detection framework for evaluating whether operational cloud seeding activities produce detectable signatures in the NEXRAD radar archive. By combining convolutional autoencoders with a temporal causal transformer, the framework successfully identified unusual radar evolution patterns across both winter orographic and warm-season convective environments, achieving AUROCs of 0.841 and 0.742 respectively for k-NN scoring and 0.9605 for the transformer on Rainmaker test sequences.

Results from the SNOWIE campaign (Tessendorf et al., 2019) suggest that physically real glaciogenic cloud seeding plume structures may exist below the detection threshold of operational NEXRAD systems, particularly within shallow winter orographic clouds dominated by small ice particles. Results from the TPWMA operational program demonstrate that anomalous convective evolution can be identified within seeded environments but also show that operational targeting introduces substantial attribution challenges because TPWMA aircraft are dispatched preferentially into already-favorable atmospheric conditions.

Overall, the study does not find clean evidence of a distinct NEXRAD-scale cloud seeding signature. However, the absence of such a signature does not demonstrate the absence of cloud seeding effects. The project clarifies the observational and methodological constraints under which radar-based cloud seeding evaluation must operate and demonstrates the potential role of unsupervised machine learning as a screening tool for identifying meteorologically unusual events within large radar archives.

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